

Homework — Week 2, Day 1

Due: **before the next session** (instructor: fill date ____).

Submit in a GitHub repo (same as `hello.go`, or a new `epic-go-week02` repo). Folder layout:

```
week02/  
  zeros.go  
  greet.go  
  names/  
    go.mod  
    names.go  
    names_test.go  
  exercises.md          # short answers for 1-9
```

No zip files, no WhatsApp screenshots-only.

Exercises 1-9 (write the answer in `exercises.md`)

Run the code. Then write **one or two sentences** plus the output. Your words.

1. **Zero values.** Declare `var n int, var s string, var ok bool` and print them with `Printf` and `%d %q %t`. What printed, and why is that better than “uninitialized”?
2. **var vs :=.** Show one `var` at package level and one `:=` inside `main`. What happens if you move the `:=` to package level?
3. **Conversion.** `year := 2026` then add `0.5`. Make it compile. What is `int(3.9)`?
4. **const.** Declare `const institute = "Epic Learn"`. Try to assign a new value. Paste the compiler error.
5. **iota (light).** A `const` block with `active, idle, archived` using `iota`. Print the three numbers.
6. **Operators.** Print `10 / 3` and `float64(10) / 3`. Explain the difference. Show `enrolled && paid` when one is `false`.
7. **fmt verbs.** Print one value with `%v`, `%T`, and `%q`. When would you pick each?
8. **strings.** `TrimSpace`, `ToUpper`, and `Contains` on `" sms "`. What does each return?
9. **Bytes vs runes.** Print `len` and `utf8.RuneCountInString` for `"café"` and `"සිංහල"`. Measure — do not guess how many “letters” Sinhala has. Why are the two numbers different?

Exercise 10 — `FullName` + `go test`

```
func FullName(first, last string) string
```

- Trim spaces on both parts.
- Join with **one** space.
- Empty last name → return only the trimmed first name (no trailing space).
- Empty first name → return only the trimmed last name.

Minimum tests in `names_test.go`:

Input	Want
<code>"Ada", "Lovelace"</code>	<code>"Ada Lovelace"</code>
<code>" Ada ", " Lovelace "</code>	<code>"Ada Lovelace"</code>
<code>"Ada", " "</code>	<code>"Ada"</code>

```
cd names
go test
```

Must be green. You may add more table rows. We are **not** mastering `testing` this week — you are seeing the command and a failing vs passing test.

How it is checked

Requirement	Pass
Repo URL sent	Yes
<code>zeros.go</code> and <code>greet.go</code> run	Shown in class or committed
<code>go test</code> in <code>names/</code> is green	Including the trim cases
<code>exercises.md</code> is your words	Oral question next class
<code>gofmt</code> clean, no binaries	<code>.gitignore</code> working